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MIS 401: Business Intelligence & Analytics  
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# **Diabetes Risk Prediction by Patient Health Indicators**

## **UCI x CDC**

MIS 401 Project Proposal - Step 1

### **Data Source**

**UCI Machine Learning Repository: CDC Diabetes Health Indicators**

<https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>

**Kaggle Dataset Homepage**

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>

This project uses the CDC Diabetes Health Indicators dataset from the publicly available, widely trusted, and well-verified UCI Machine Learning Repository. UCI identifies the dataset as externally linked, referencing the CDC BRFSS as the underlying source and providing a Dataset Home Page link for the cleaned dataset.

### **Data Description**

The dataset includes healthcare statistics and the lifestyle survey information from collected sources in the United States to improve our understanding of the relationship between lifestyle and diabetes. The dataset, classifying multi-class diabetes-related outcomes, contains 253,680 observations, 21 features, and no missing values. The scope of this repository is well-suited for predictive modeling and analysis.

### **Variables**

The dataset covers health, lifestyle, and demographic variables that may reasonably help explain diabetes risk, with variables including high blood pressure, high cholesterol, cholesterol check history, body mass index (BMI), smoking history, stroke history, heart disease or heart attack history, physical activity, fruit consumption, vegetable consumption, heavy alcohol consumption, healthcare access, inability to see

a doctor because of cost, general health, mental health, physical health, difficulty walking, sex, age, education, and income. These variables will be used to identify which factors have the strongest association with diabetes risk.

## **Response Variable**

The original dataset response variable is Diabetes\_012, a categorical variable with three classes: 0 = no diabetes, 1 = prediabetes, and 2 = diabetes. The meaningful relationships I expect from the variables listed with the response variable are age, BMI, high blood pressure, heart disease history, general health, and physical activity.

## **Methods**

I plan to apply classification methods to compare model performance, such as logistic regression, k-nearest neighbors (KNN), and decision trees. I will evaluate how these approaches differ across the three categorical classes of no diabetes, prediabetes, and diabetes, as they provide sufficient and effective classification analysis.

## **Why We Care**

Diabetes is a major public health issue, particularly in the United States. Identifying relationships between health, lifestyle, and demography can aid the development of future intervention and prevention strategies. Predictive models can help healthcare organizations identify risk patterns, improve decision-making, and reallocate resources to patient care groups. This topic is also personally meaningful to me because Type II diabetes has affected my father. Last summer, I worked for a biopharmaceutical company in Commercial Operations, Business Intelligence, and AI Solutions Analysis. This gives me motivation to deepen my perspective on how analytics can be leveraged and aligned to support healthcare insight and action.

## **AI Disclosure Statement**

Generative AI organized and polished the proposal's skeletal structure. The dataset selection, project direction, methodology, and analytical decisions are my own.